# 2024, Mar. 2024 Release LOC:Acute Adult
# Non-Traumatic Bleeding

**Utilization Benchmarks:** *Multiple options available, see table below for options*





## Overview

**Instruction:**
This subset is for patients with bleeding (hemoptysis, hemophilia, epistaxis, gross hematuria, spontaneous hematoma, uterine bleeding, and vaginal bleeding). Excluded from this subset are bleeding from traumatic injury and microscopic hematuria. For hemoptysis due to infection, including pneumonia or lung abscess, use the Infection: Pneumonia subset. For gastrointestinal bleeding use the Gastrointestinal (GI) Bleeding subset. For a pulmonary embolus confirmed by imaging use the Pulmonary Embolism subset.
**Introduction:**
Bleeding refers to the loss of blood. This may occur internally when blood leaks from vessels or organs, or externally when blood is lost through natural openings (e.g., vagina, mouth, rectum) or through breaks in the skin. Bleeding may be subtle and lead to anemia, or it may be in the form of a serious hemorrhage. Determination of the source of bleeding and control is essential for treatment.
**Evaluation and Treatment:**
Evaluation of bleeding includes identifying the source through diagnostic testing, which may include endoscopy, bronchoscopy, colonoscopy, ultrasound, angiography, nuclear medicine scan, computed tomography, urography, or cystography. Treatment is initially directed at stopping the bleeding, providing intravenous fluid and/or blood product transfusion as needed, monitoring of hemoglobin and hematocrit, and evaluation of coagulation studies to rule out a bleeding disorder, if indicated (Borhart, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:263-7; Deshwal et al., Semin Respir Crit Care Med 2021, 42: 145-59; Tebo et al., J Emerg Med 2020, 58: 756-66; Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38; Carson et al., JAMA 2016, 316: 2025-35).

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, Agency for Healthcare Research and Quality (AHRQ) Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The content is based on a variety of references which are cited at specific criteria points throughout the subset.

(Excludes PO medications unless noted)

**Select Day, One:**

- ⊖ **Initial review, One:** (1, 2)
  - (From arrival in ED through decision to admit)
  - (Symptom or finding within 24h)
  - ⊖ **OBSERVATION, ≥ One:** (3)
    - ⊖ Epistaxis, anterior and, **All:** (4, 5)
      - Bleeding through packing (6)
      - Hct or Hb monitoring at least 2x/24h
      - Oximetry
    - ⊖ Hematuria and, ≥ **One:** (7, 8)
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (9)
      - Exertional dyspnea worse than baseline (10)
      - ⊖ Orthostatic hypotension, **Both:** (11)
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
      - Syncope (12)
      - Urinary retention due to blood clot (13)
    - ⊖ Uterine or vaginal bleeding, abnormal (excludes pregnancy) and, ≥ **One:** (14, 15)
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (9)
      - ≥ 3 pads/h (16)
      - Exertional dyspnea worse than baseline (10)
      - INR ≥ 2.0
      - ⊖ Orthostatic hypotension, **Both:** (11)
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
      - Syncope (12)
  - ⊖ **ACUTE, ≥ One:** (17)
    - ⊖ Epistaxis, anterior, persistent and, **Both:** (4, 5)
      - Embolization or arterial ligation/cautery performed within 24h
      - Hct or Hb monitoring at least 2x/24h and blood product transfusion
    - ⊖ Epistaxis, posterior and, ≥ **One:** (4, 5)
      - Embolization or arterial ligation/cautery performed within 24h
      - Posterior packing (18, 19)
    - ⊖ Hemophilia or von Willebrand disease and, **One:** (20, 21)
      - Persistent bleeding with hemodynamic stability (22)
    - ⊖ Hemoptysis and, ≥ **One:** (23)
      - Dyspnea
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (24)
      - $O_2$ sat < 90%(0.90) and < baseline (10)
    - ⊖ Spontaneous hematoma and, **Both:** (25)
      - Hemodynamic stability (22)
      - ⊖ Intervention, ≥ **One:**
        - Blood product transfusion
        - Hct or Hb monitoring at least 2x/24h
        - Repeat imaging
        - Reversal of anticoagulant
  - ⊖ **CRITICAL, ≥ One:** (26)
    - Embolization performed within 24h (27)
    - ⊖ Hemoptysis and, ≥ **One:** (28, 29)
      - ⊖ Bronchoscopy and, **One:** (30)
        - Balloon tamponade
        - Cold saline lavage

            - Instillation of vasoconstrictive agent
            - Topical hemostatic tamponade [31]
        - Bronchial artery embolization (BAE)
    - Hemodynamic instability, **All:** [32]
        - ED treatment, ≥ **One:**
            - ≥ 2 units blood product transfusion
            - ≥ 2 liters IV fluid [33]
            - ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
        - Findings after ED treatment, ≥ **One:**
            - Heart rate > 120/min
            - Systolic blood pressure < 100 mmHg and < baseline
            - MAP < 65 mmHg
        - Requiring ongoing resuscitation
    - Hemodynamic monitoring, invasive [34]
    - Hemophilia or von Willebrand disease and, **One:** [35]
        - Evidence of hemorrhage on CT or MRI and, ≥ **One:** [36]
            - Compartment syndrome
            - Hemodynamic instability
            - Mental status change [37]
            - New onset neurological deficit, ≥ **One:**
                - Alteration in speech [38]
                - Gait disturbance
                - Paresis or paralysis [39]
                - Sensory deficit [40]
                - Visual impairment [41]
    - HFNC [42, 43]
    - Mechanical ventilation [44]
    - NIPPV [45]
    - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 1, One:** [1]
  (Symptom or finding within 24h)
    - **OBSERVATION, ≥ One:** [3]
        - Epistaxis, anterior and, **All:** [4, 5]
            - Bleeding through packing [6]
            - Hct or Hb monitoring at least 2x/24h
            - Oximetry
        - Hematuria and, **Both:** [7, 8]
            - Finding, ≥ **One:**
                - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [9]
                - Exertional dyspnea worse than baseline [10]
                - Orthostatic hypotension, **Both:** [11]
                    - Sustained drop in blood pressure within 3 min of sitting or standing
                    - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
                - Syncope [12]
                - Urinary retention due to blood clot [13]
            - Intervention, ≥ **One:**
                - Bladder irrigation, continuous or cyclic ≤ 2d
                - Blood product transfusion
                - CT urography ≤ 24h
                - Cystoscopy ≤ 24h
        - Uterine or vaginal bleeding, abnormal (excludes pregnancy) and, **Both:** [14, 15]
            - Finding, ≥ **One:**
                - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [9]

- ≥ 3 pads/h [16]
- Exertional dyspnea worse than baseline [10]
- INR ≥ 2.0
- Orthostatic hypotension, **Both:** [11]
  - Sustained drop in blood pressure within 3 min of sitting or standing
  - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
- Syncope [12]

- Hct or Hb monitoring at least 2x/24h and, **All:**
  - Blood product transfusion or IV fluid [46]
  - Hormone therapy administered or contraindicated [47, 48]
  - Procedure, ≥ **One:**
    - Transvaginal or pelvic ultrasound performed within 24h
    - Dilation and curettage (D&C) performed within 24h

- **ACUTE,** ≥ **One:** [17]
  - Epistaxis, anterior, persistent and, **Both:** [4, 5]
    - Embolization or arterial ligation/cautery performed within 24h
    - Hct or Hb monitoring at least 2x/24h and blood product transfusion
  - Epistaxis, posterior and, ≥ **One:** [4, 5]
    - Embolization or arterial ligation/cautery performed within 24h
    - Posterior packing [18, 19]
  - Hemophilia or von Willebrand disease and, **Both:** [20, 21]
    - Persistent bleeding with hemodynamic stability [22]
    - Intervention, ≥ **One:**
      - Cryoprecipitate
      - Desmopressin [49]
      - Factor replacement
      - Fresh frozen plasma (FFP)
  - Hemoptysis and, **Both:** [23]
    - Finding, ≥ **One:**
      - Dyspnea
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
      - $O_2$ sat < 90%(0.90) and < baseline [10]
    - Intervention, ≥ **One:**
      - Bronchoscopy performed within 24h
      - CT or computed tomography angiography (CTA) of chest performed within 24h
      - Requiring supplemental oxygen, new or > baseline requirement [10]
  - Spontaneous hematoma and, **Both:** [25]
    - Hemodynamic stability [22]
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - Hct or Hb monitoring at least 2x/24h
      - Repeat imaging
      - Reversal of anticoagulant
- **CRITICAL,** ≥ **One:** [26]
  - Embolization performed within 24h [27]
  - Hemoptysis and, ≥ **One:** [28, 29]
    - Bronchoscopy and, **One:** [30]
      - Balloon tamponade
      - Cold saline lavage
      - Instillation of vasoconstrictive agent
      - Topical hemostatic tamponade [31]
    - Bronchial artery embolization (BAE)
  - Hemodynamic instability, **All:** [32]

- ⊖ ED treatment, ≥ **One:**
  - − ≥ 2 units blood product transfusion
  - − ≥ 2 liters IV fluid [33]
  - − ≥ 1 unit blood product transfusion and ≥ 1 liter IV fluid
- ⊖ Findings after ED treatment, ≥ **One:**
  - − Heart rate > 120/min
  - − Systolic blood pressure < 100 mmHg and < baseline
  - − MAP < 65 mmHg
- − Requiring ongoing resuscitation

− Hemodynamic monitoring, invasive [34]

⊖ Hemophilia or von Willebrand disease and, **Both:** [35]

- ⊖ Evidence of hemorrhage on CT or MRI and, ≥ **One:** [36]
  - − Compartment syndrome
  - − Hemodynamic instability
  - − Mental status change [37]
  - ⊖ New onset neurological deficit, ≥ **One:**
    - − Alteration in speech [38]
    - − Gait disturbance
    - − Paresis or paralysis [39]
    - − Sensory deficit [40]
    - − Visual impairment [41]
- ⊖ Intervention, ≥ **One:**
  - − Cryoprecipitate
  - − Factor replacement
  - − Fresh frozen plasma (FFP)

− HFNC [42, 43]

− Mechanical ventilation [44]

− NIPPV [45]

− Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 2, One:** [50]

⊖ **OBSERVATION, One:**

- ⊖ Epistaxis, anterior and, **One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
    - − Bleeding controlled or resolved
    - − Hct or Hb within acceptable limits [52]
    - − Vital signs within acceptable limits [52]
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
    - − For a condition other than epistaxis use appropriate subset (Episode Day 1) based on clinical findings
    - − Refer for secondary review
- ⊖ Hematuria and, **One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
    - − Bleeding controlled or resolved
    - − Hct or Hb within acceptable limits [52]
    - − Vital signs within acceptable limits [52]
  - ⊖ **Non-responder**, not clinically stable for discharge and exceeds initial 24h Observation for this condition, **One:**
    - − For hematuria use Episode Day 2 at a higher level of care
    - − For a condition other than hematuria use appropriate subset (Episode Day 1) based on clinical findings
    - − Refer for secondary review
- ⊖ Uterine or vaginal bleeding, abnormal (excludes pregnancy) and, **One:** [14, 15]
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]

- − Bleeding controlled or resolved
- − Hct or Hb within acceptable limits (52)
- − Vital signs within acceptable limits (52)

- ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, **Both:**
  - ⊖ Finding, ≥ **One:**
    - − Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (9)
    - − ≥ 3 pads/h (16)
    - − Exertional dyspnea worse than baseline (10)
    - − INR ≥ 2.0
    - ⊖ Orthostatic hypotension, **Both:** (11)
      - − Sustained drop in blood pressure within 3 min of sitting or standing
      - − Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
    - − Syncope (12)
  - ⊖ Hct or Hb monitoring at least 2x/24h and, **All:**
    - − Blood product transfusion or IV fluid (46)
    - − Hormone therapy administered or contraindicated (47, 48)
    - ⊖ Procedure, ≥ **One:**
      - − Transvaginal or pelvic ultrasound performed within 24h
      - − Dilation and curettage (D&C) performed within 24h

- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** (51)
    - − Bleeding controlled or resolved
    - − Hct or Hb within acceptable limits (52)
    - − $O_2$ sat within acceptable limits (52)
    - − Vital signs within acceptable limits (52)
    - − No complication or active comorbidity relevant to this episode of care (53)
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - ⊖ Epistaxis, posterior and, ≥ **One:** (4, 5)
      - − Embolization or arterial ligation/cautery performed within 24h
      - − Hct or Hb monitoring at least daily ≤ 24h
      - − Posterior packing (18, 19)
    - ⊖ Hematuria and, **Both:**
      - ⊖ Finding, ≥ **One:**
        - − Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) (9)
        - − Hct or Hb decreased last 24h
        - − Exertional dyspnea worse than baseline (10)
        - − Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 (24)
        - ⊖ Orthostatic hypotension, **Both:** (11)
          - − Sustained drop in blood pressure within 3 min of sitting or standing
          - − Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
        - − Persistent hematuria or blood clot
        - − Syncope (12)
      - ⊖ Intervention, ≥ **One:**
        - − Bladder irrigation, continuous or cyclic ≤ 2d
        - − Blood product transfusion
    - ⊖ Hemophilia or von Willebrand disease and, **Both:**
      - − Persistent bleeding with hemodynamic stability (22)
      - ⊖ Intervention, ≥ **One:**
        - − Cryoprecipitate
        - − Factor replacement
        - − Fresh frozen plasma (FFP)
    - ⊖ Hemoptysis and, **Both:** (23)
      - ⊖ Finding, ≥ **One:**

- Dyspnea
- Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
- $O_2$ sat < 90%(0.90) and < baseline [10]

⊖ Intervention, ≥ **One:**
- Bronchoscopy performed within last 24h
- CT or computed tomography angiography (CTA) of chest performed within last 24h
- Requiring supplemental oxygen, new or > baseline requirement [10]

⊖ Spontaneous hematoma and, **Both:** [25]
- Hemodynamic stability [22]

⊖ Intervention, ≥ **One:**
- Blood product transfusion
- Hct or Hb monitoring at least 2x/24h
- Repeat imaging [54]

- Post Critical care ≤ 24h

⊖ **INTERMEDIATE, ≥ One:** [55]
- HFNC and respiratory monitoring every 3-4h [56, 57]
- NIPPV [45]
- Oxygen ≥ 40%(0.40), ≤ 2d [42]

⊖ **CRITICAL, ≥ One:**
- Embolization performed within 24h [27]
- Post embolization ≤ 24h [58]

⊖ Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [32]

⊖ Finding, ≥ **One:**
- Heart rate > 120/min
- Systolic blood pressure < 100 mmHg and < baseline
- MAP < 65 mmHg

⊖ Intervention, ≥ **One:**
- Blood product transfusion
- IV fluid resuscitation ≥ 1L ≤ 24h [33]

- Hemodynamic monitoring, invasive [34]
- HFNC and respiratory monitoring every 1-2h [42, 43, 57]
- Mechanical ventilation

⊖ NIPPV and, ≥ **One:** [45]
- $FiO_2$ > 50%(0.50)
- Respiratory intervention every 1-2h [59]

- Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 3, One:** [60]

⊖ **OBSERVATION, One:**

⊖ Uterine or vaginal bleeding, abnormal (excludes pregnancy) and, **One:** [14, 15]

⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
- Bleeding controlled or resolved
- Hct or Hb within acceptable limits [52]
- Vital signs within acceptable limits [52]

⊖ **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One:**
- For uterine or vaginal bleeding use Episode Day 3 at a higher level of care
- For a condition other than uterine or vaginal bleeding use appropriate subset (Episode Day 1) based on clinical findings
- Refer for secondary review

⊖ **ACUTE, One:**

⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
- Bleeding controlled or resolved

- Hct or Hb within acceptable limits [52]
- $O_2$ sat within acceptable limits [52]
- Vital signs within acceptable limits [52]
- No complication or active comorbidity relevant to this episode of care [53]

- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
  - Epistaxis, posterior and, ≥ **One:** [4, 5]
    - Embolization or arterial ligation/cautery performed within 24h
    - Hct or Hb monitoring at least daily ≤ 24h
    - Posterior packing [18, 19]
  - Hematuria and, **Both:**
    - Finding, ≥ **One:**
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [9]
      - Hct or Hb decreased last 24h
      - Exertional dyspnea worse than baseline [10]
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
      - Orthostatic hypotension, **Both:** [11]
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
      - Persistent hematuria or blood clot
      - Syncope [12]
    - Intervention, ≥ **One:**
      - Bladder irrigation, continuous or cyclic ≤ 2d
      - Blood product transfusion
  - Hemophilia or von Willebrand disease and, **Both:**
    - Persistent bleeding with hemodynamic stability [22]
    - Intervention, ≥ **One:**
      - Cryoprecipitate
      - Factor replacement
      - Fresh frozen plasma (FFP)
  - Hemoptysis and, **Both:** [23]
    - Finding, ≥ **One:**
      - Dyspnea
      - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
      - $O_2$ sat < 90%(0.90) and < baseline [10]
    - Requiring supplemental oxygen, new or > baseline requirement [10]
  - Spontaneous hematoma and, **Both:** [25]
    - Hemodynamic stability [22]
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - Hct or Hb monitoring at least 2x/24h
      - Repeat imaging [54]
  - Uterine or vaginal bleeding, abnormal (excludes pregnancy) and, **Both:** [14, 15]
    - Finding, ≥ **One:**
      - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [9]
      - ≥ 3 pads/h [16]
      - Exertional dyspnea worse than baseline [10]
      - Orthostatic hypotension, **Both:** [11]
        - Sustained drop in blood pressure within 3 min of sitting or standing
        - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
      - Syncope [12]
    - Intervention, **Both:**
      - Blood product transfusion or IV fluid [46]
      - Hct or Hb monitoring at least 2x/24h

- Post Critical care ≤ 24h
- ⊖ **INTERMEDIATE,** ≥ **One:** [55]
  - HFNC and respiratory monitoring every 3-4h [56, 57]
  - NIPPV [45]
  - Oxygen ≥ 40%(0.40), ≤ 2d [42]
- ⊖ **CRITICAL,** ≥ **One:**
  - Embolization performed within 24h [27]
  - Post embolization ≤ 24h [58]
  - ⊖ Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [32]
    - ⊖ Finding, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - ⊖ Intervention, ≥ **One:**
      - Blood product transfusion
      - IV fluid resuscitation ≥ 1L ≤ 24h [33]
  - Hemodynamic monitoring, invasive [34]
  - HFNC and respiratory monitoring every 1-2h [42, 43, 57]
  - Mechanical ventilation
  - ⊖ NIPPV and, ≥ **One:** [45]
    - $FiO_2$ > 50%(0.50)
    - Respiratory intervention every 1-2h [59]
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

⊖ **Episode Day 4, One:**
- ⊖ **ACUTE, One:**
  - ⊖ **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
    - Bleeding controlled or resolved
    - Hct or Hb within acceptable limits [52]
    - $O_2$ sat within acceptable limits [52]
    - Vital signs within acceptable limits [52]
    - No complication or active comorbidity relevant to this episode of care [53]
  - ⊖ **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
    - ⊖ Epistaxis, posterior and, ≥ **One:** [4, 5]
      - Embolization or arterial ligation/cautery performed within 24h
      - Hct or Hb monitoring at least daily ≤ 24h
      - Posterior packing [18, 19]
    - ⊖ Hematuria and, **Both:**
      - ⊖ Finding, ≥ **One:**
        - Hct < 21%(0.21) or Hb < 7.0 g/dL(70 g/L) [9]
        - Hct or Hb decreased last 24h
        - Exertional dyspnea worse than baseline [10]
        - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
        - ⊖ Orthostatic hypotension, **Both:** [11]
          - Sustained drop in blood pressure within 3 min of sitting or standing
          - Systolic drop ≥ 20 mmHg or diastolic drop ≥ 10 mmHg
        - Persistent hematuria or blood clot
        - Syncope [12]
      - ⊖ Intervention, ≥ **One:**
        - Bladder irrigation, continuous or cyclic ≤ 2d
        - Blood product transfusion
    - ⊖ Hemophilia or von Willebrand disease and, **Both:**
      - Persistent bleeding with hemodynamic stability [22]

- Intervention, ≥ **One:**
  - Cryoprecipitate
  - Factor replacement
  - Fresh frozen plasma (FFP)
- Hemoptysis and, **Both:** [23]
  - Finding, ≥ **One:**
    - Dyspnea
    - Mental status changes (excludes coma, stupor, or obtundation) or GCS 9-14 [24]
    - $O_2$ sat < 90%(0.90) and < baseline [10]
  - Requiring supplemental oxygen, new or > baseline requirement [10]
- Post Critical care ≤ 24h

- **INTERMEDIATE, ≥ One:** [55]
  - HFNC and respiratory monitoring every 3-4h [56, 57]
  - NIPPV [45]
  - Oxygen ≥ 40%(0.40), ≤ 2d [42]
- **CRITICAL, ≥ One:**
  - Embolization performed within 24h [27]
  - Post embolization ≤ 24h [58]
  - Hemodynamic instability persisting after ≥ 1L of IVF resuscitation or ≥ 1 unit blood product transfusion and, **Both:** [32]
    - Finding, ≥ **One:**
      - Heart rate > 120/min
      - Systolic blood pressure < 100 mmHg and < baseline
      - MAP < 65 mmHg
    - Intervention, ≥ **One:**
      - Blood product transfusion
      - IV fluid resuscitation ≥ 1L ≤ 24h [33]
  - Hemodynamic monitoring, invasive [34]
  - HFNC and respiratory monitoring every 1-2h [42, 43, 57]
  - Mechanical ventilation
  - NIPPV and, ≥ **One:** [45]
    - $FiO_2$ > 50%(0.50)
    - Respiratory intervention every 1-2h [59]
  - Vasopressor or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

- **Episode Day 5, One:**
  - **ACUTE, One:**
    - **Responder**, discharge expected today if clinically stable last 12h, **All:** [51]
      - Bleeding controlled or resolved
      - Hct or Hb within acceptable limits [52]
      - $O_2$ sat within acceptable limits [52]
      - Vital signs within acceptable limits [52]
      - No complication or active comorbidity relevant to this episode of care [53]
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **INTERMEDIATE, One:**
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One:**
      - **Use Extended Stay subset**
      - Refer for secondary review
  - **CRITICAL, One:**
    - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One:**

- Use Extended Stay subset
- Refer for secondary review

---

**Discharge, Level of Care, One:** [61]

- **HOME, Both:**
  - Level of care appropriateness, **Both:**
    - Home environment safe and accessible [62]
    - Patient or caregiver demonstrates ability to manage care
  - Bleeding discharge planning, **All:**
    - Provide comprehensive written discharge and teaching instructions [63]
    - Education on the importance of, **All:**
      - Follow-up appointments
      - Avoiding excessive alcohol if indicated
      - Knowing when to seek medical care
      - Obtaining flu and pneumococcal vaccinations
    - Medication reconciliation [64]
    - Follow-up care planned [65]
    - Identify and address transportation needs
- **HOME CARE, One:**
  - Level of care appropriateness, **All:**
    - Home environment safe and accessible [62]
    - Patient or caregiver willing and able to learn care
    - Outpatient management contraindicated or unavailable [66, 67]
    - Skilled services, ≥ **One:** [68]
      - Skilled nursing [69]
      - Skilled therapy: PT, OT, or ST
      - Behavioral health [70]
    - Bleeding discharge planning,
    - Provide comprehensive written discharge and teaching instructions [63]
    - Education on the importance of, **All:**
      - Follow-up appointments
      - Avoiding excessive alcohol if indicated
      - Knowing when to seek medical care
      - Obtaining flu and pneumococcal vaccinations
    - Medication reconciliation [64]
    - Follow-up care planned [65]
    - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both:**
  - Level of care appropriateness, **One:**
    - Support systems available [71]
  - Skilled treatment, ≥ **One:**
    - Clinical assessment [72]
    - Individual, group, or family counseling
    - Psychiatric, substance, or medication evaluation [73]
- **SKILLED MEDICAL OR THERAPY, Both:**
  - Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
    - Treatment precluded in a lower level
    - Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 1-2x/24h [69]
      - Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [74, 75]
        - Functional impairments requiring at least supervision [76]

---

- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk
- ⊖ Complete prior to facility transfer, **All:**
  - Medication reconciliation [64]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

- ⊖ **SUBACUTE MEDICAL OR THERAPY, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, ≥ **One:**
      - Medical and skilled nursing interventions or assessment 4h/24h [69]
      - ⊖ Therapy, **All:**
        - Goal directed therapy with rehabilitation potential [74, 75]
        - ≥ 2 functional impairments requiring at least minimum assistance [76]
        - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation [64]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **SUBACUTE REHAB, Both:**
  - ⊖ Level of care appropriateness, **All:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - Treatment precluded in a lower level
    - ⊖ Services, **All:**
      - Goal directed therapy with rehabilitation potential [74, 75]
      - ≥ 2 functional impairments requiring at least minimum assistance [76]
      - Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation [64]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **ACUTE REHAB, Both:**
  - ⊖ Level of care appropriateness, **Both:**
    - Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
    - ⊖ Services, **All:**
      - Goal directed therapy with rehabilitation potential [74, 75]
      - ≥ 2 functional impairments requiring at least minimum assistance [76]
      - Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines [77]
  - ⊖ Complete prior to facility transfer, **All:**
    - Medication reconciliation [64]
    - Obtain and complete forms for facility
    - Obtain discharge summary and transmit to facility and medical practitioner
    - Arrange transportation
- ⊖ **LONG TERM ACUTE CARE, Both:**
  - ⊖ Level of care appropriateness, **Both:**
    - Treatment precluded in a lower level
    - ⊖ In lieu of acute or continued hospitalization or failed lower level of care, ≥ **One:**
      - Primary condition or illness and treatment of active comorbid condition(s) [78]
      - Ventilator dependent ≥ 6h/d and weaning planned [79, 80]
      - Acute and comorbid conditions requiring prolonged hospitalization [81]

- Complete prior to facility transfer, **All:**
  - Medication reconciliation [64]
  - Obtain and complete forms for facility
  - Obtain discharge summary and transmit to facility and medical practitioner
  - Arrange transportation

**Utilization Benchmarks**

| Condition or Procedure | % Pd Obs | LOS (days) | Type |
|---|---|---|---|
| **150 EPISTAXIS WITH MCC** | **n/a** | **3.4** | **CMS GMLOS** |
| **151 EPISTAXIS WITHOUT MCC** | **n/a** | **2.2** | **CMS GMLOS** |
| **695 KIDNEY AND URINARY TRACT SIGNS AND SYMPTOMS WITH MCC** | **n/a** | **3.7** | **CMS GMLOS** |
| **696 KIDNEY AND URINARY TRACT SIGNS AND SYMPTOMS WITHOUT MCC** | **n/a** | **2.3** | **CMS GMLOS** |
| **698 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITH MCC** | **n/a** | **4.9** | **CMS GMLOS** |
| **699 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITH CC** | **n/a** | **3.3** | **CMS GMLOS** |
| **700 OTHER KIDNEY AND URINARY TRACT DIAGNOSES WITHOUT CC/MCC** | **n/a** | **2.3** | **CMS GMLOS** |
| **760 MENSTRUAL AND OTHER FEMALE REPRODUCTIVE SYSTEM DISORDERS WITH CC/MCC** | **n/a** | **2.8** | **CMS GMLOS** |
| **761 MENSTRUAL AND OTHER FEMALE REPRODUCTIVE SYSTEM DISORDERS WITHOUT CC/MCC** | **n/a** | **1.8** | **CMS GMLOS** |
| **813 COAGULATION DISORDERS** | **n/a** | **3.7** | **CMS GMLOS** |
| **EPISTAXIS** | **61%** | **3** | **InterQual** |
| **HEMATURIA** | **41%** | **3.8** | **InterQual** |
| **HEMOPHILIA** | **13%** | **4.4** | **InterQual** |
| **HEMOPTYSIS** | **53%** | **3.2** | **InterQual** |
| **VAGINAL BLEEDING** | **63%** | **1.9** | **InterQual** |

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

## Notes:

**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.
For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.

- **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

**NOTE:** In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

**NOTE:** CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**2:**

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.
Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.
**Note**: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.
***For further clarification, see the Acute Level of Care Review Process.***

**3:**
Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

**4:**
Initial management of epistaxis may include combinations of direct compression, topical application of vasoconstrictors, chemical or electrical cautery, and nasal packing. Patients unresponsive to initial local treatment may require intensive management (e.g., surgical ligation/cautery, endovascular embolization) (Seikaly, N Engl J Med 2021, 384: 944-51; Tran et al., Open Access Emerg Med 2021, 13: 527-33; Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38).

**5:**
Severity of nasal bleeding can be determined in several ways (Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38):

- Bleeding > 30 minutes in a 24h period
- History of hospitalization for prior nosebleed
- History of blood transfusion for prior nosebleed
- More than 3 recent episodes of nasal bleeding

Additionally, the presence of hypertension, cardiopulmonary disease, anemia, bleeding disorders, and liver or kidney disease may assist in determining who requires prompt management (Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38; Krulewitz and Fix, Emerg Med Clin North Am 2019, 37: 29-39).

**6:**
**Instruction:** Application of this criteria point is appropriate for patients with failed emergency department (ED) management (e.g., patients who were discharged from the ED and have returned due to continued bleeding despite ED treatment).

**7:**
**Instruction:** These criteria are intended for the evaluation and treatment of gross hematuria and exclude microscopic hematuria. Macroscopic or gross hematuria is hematuria that is visible to the naked eye.

**8:**
In adults, hematuria is most frequently the result of nephrolithiasis, pyelonephritis, benign prostatic hypertrophy, or malignancy. In rare circumstances hematuria can be associated with a life-threatening vascular event. Evaluation may include a history and physical examination, laboratory studies, and imaging, if indicated. Most patients with hematuria can be discharged home with outpatient follow-up, however, hematuria resulting in hemodynamic instability, urinary obstruction, or suspected vascular etiology requiring angiographic imaging, warrants urology consultation and/or admission (Willis and Tewelde, Emerg Med Clin North Am 2019, 37: 755-69).

**9:**
Blood transfusion practice guidelines are intended to improve clinical outcomes while avoiding unnecessary transfusions that may not be clinically indicated. Recent high-quality evidence has resulted in guideline updates recommending lower hemoglobin transfusion thresholds for most hospitalized adult and pediatric patients. While individual institutional practice patterns may vary, the American Association of Blood Banks (AABB) recommendation for hemodynamically stable patients, including critically ill patients, is a restrictive transfusion threshold for a hemoglobin of less than 7 g/dL for most patients. A hemoglobin threshold of 7.5 g/dL is

recommended for transfusion if patients are undergoing cardiac surgery and a threshold of 8 g/dL if patients are undergoing orthopedic surgery or have pre-existing cardiac disease. There are exceptions to these parameters including patients with acute coronary syndrome, severe thrombocytopenia, and chronic transfusion-dependent anemia (Carson et al., Jama 2023: E1-E11; Chegondi et al., Transfus Med Hemother 2016, 43: 297-301).

**10:**

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

**11:**

Orthostatic hypotension (Juraschek et al., JAMA Intern Med 2017, 177: 1316-23).

**12:**

Syncope is the transient loss of consciousness caused by diminished cerebral blood flow, identified as brief, with spontaneous onset and recovery (Brignole et al., Eur Heart J 2018, 39: 1883-948; Shen et al., Journal of the American College of Cardiology 2017, 70: e39-e110).

**13:**

**Instruction:** Application of this criteria point is appropriate for patients with or without a Foley catheter in place.

**14:**

The American College of Obstetricians and Gynecologists defines abnormal uterine bleeding as a change in bleeding "regularity, volume, frequency, or duration" in nonpregnant individuals (Obstet Gynecol 2013, 121: 891-6. Reaffirmed 2020; Obstet Gynecol 2012, 120: 197-206).

**15:**

Confirming pregnancy status is the most important determination to make when evaluating patients with uterine bleeding (Borhart, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:263-7; Dyne and Miller, Emerg Med Clin North Am 2019, 37: 153-64). Patients with abnormal uterine bleeding who are experiencing signs or symptoms, are anticoagulated, or have bleeding caused by a malignancy should be managed in Observation status for monitoring and treatment (Wheatley, Emerg Med Clin North Am 2017, 35: 701-12). Patients with brisk active bleeding and hemodynamic instability should be managed at the Critical level of care.

**16:**

There is a wide array of menstrual pads ranging from "light" to "maxi" to "heavy" pads. Although the number of pads utilized per hour is a common qualifier of menstrual bleeding, it is recommended that patients be questioned regarding the quality of their bleeding (i.e., light, normal, heavy), the impact on their life, and change from baseline (Dyne and Miller, Emerg Med Clin North Am 2019, 37: 153-64).

**17:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

**18:**

Nasal packing for posterior epistaxis to achieve hemostasis may be accomplished with the use of gauze, a Foley catheter, a nasal sponge/tampon, or inflatable nasal balloon catheter. Although it is an effective treatment to manage bleeding, packing the posterior nasal cavity is uncomfortable and may require sedation. Additionally, the presence of posterior packing can potentially cause reflex bradyarrhythmia or airway obstruction. Supplemental oxygen should be considered for these patients to avoid hypoxemia (Seikaly, N Engl J Med 2021, 384: 944-51; Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38).

**19:**

**Instruction:** Patients with posterior epistaxis must have their packing removed prior to discharge (Tunkel et al., Otolaryngol Head Neck Surg 2020, 162: S1-S38).

**20:**

Patients with hemophilia most commonly present with spontaneous bleeding into their joints (e.g., ankle, elbow, knee), which can occur repeatedly into the same joint in those with severe disease. Bleeding occurring into joints or muscles may potentially cause pain, nerve infringement, and decreased range of motion. Hemorrhaging may potentially become life-threatening if bleeding occurs in large muscles (e.g., iliopsoas), intestines, or intracerebrally or limb-threatening if compartment syndrome develops (Tebo et al., J Emerg Med 2020, 58: 756-66).

**21:**

Only the most experienced personnel should attempt to obtain intravenous access utilizing the smallest possible gauge catheter, and pressure should be applied over all venipuncture sites. Anticoagulants and medications with antiplatelet effects (e.g., aspirin, nonsteroidal anti-inflammatory drugs (NSAIDs)) should be avoided, as should intramuscular injections (Alblaihed et al., Am J Emerg Med 2022, 56: 21-7; Tebo et al., J Emerg Med 2020, 58: 756-66).

**22:**

Hemodynamic stability is evidenced by adequate tissue and organ perfusion. Blood pressure and heart rate are parameters commonly used to assess hemodynamic stability. Clinical findings associated with hemodynamic stability include normal peripheral pulses, warm extremities, adequate urine output, and neurological stability. The Critical level of care is appropriate for patients who develop hemodynamic instability.

**23:**

Hemoptysis is the expectoration of blood or bloody sputum that originates from the lower respiratory tract. It can range from blood-tinged sputum to gross blood. If the rate of bleeding is 100 mL per hour or greater than 100 mL in 24 hours, it is generally considered massive hemoptysis and requires urgent management (Brown, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:189-93).

**24:**

Mental status change may include confusion, disorientation, delirium, or increasing lethargy (Lacey et al., Ann Med 2019, 51: 232-51).
**Instruction:** Patients with acute coma, stupor, or obtundation should be reviewed at a higher level of care due to the need for more frequent neurological monitoring. These criteria exclude chronic coma, stupor, or obtundation (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

**25:**

Spontaneous hematoma may include:

- Intraperitoneal hematoma
- Rectus sheath hematoma
- Retroperitoneal hematoma

**26:**

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

**27:**

**Instruction:** Application of this criteria point is appropriate for patients with spontaneous hematoma (e.g., retroperitoneal hematoma, intraperitoneal hematoma, rectus sheath hematoma).

**28:**

Patients with massive hemoptysis can quickly deteriorate to respiratory distress leading to a low threshold for intubation to protect the airway. These patients are more likely to die of asphyxiation than from exsanguination. Methods of protecting the airway include keeping the patient in a decubitus position with the bleeding lung downward and selective intubation of the non-bleeding lung (Brown, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:189-93; O'Gurek and Choi, Am Fam Physician 2022, 105: 144-51; Deshwal et al., Semin Respir Crit Care Med 2021, 42: 145-59).

**29:**

Massive hemoptysis is a term used to describe a large amount of bleeding which may potentially become life-threatening. While the amount of blood loss constituting massive hemoptysis varies in the literature, the generally accepted definition is blood loss greater than 100 mL in 24 hours or a rate of 100 mL per hour or more. However, not only the volume of blood loss, which is difficult to quantify, but the rate of bleeding, the timeliness of treatment, and the patient's underlying health condition(s) should be considered. Patients should be managed at the Critical level of care when there is significant bleeding causing airway obstruction requiring intubation, mechanical ventilation, severe hypoxemia, or severe hypotension (Brown, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:189-93; O'Gurek and Choi, Am Fam Physician 2022, 105: 144-51; Prey et al., Surg Clin North Am 2022, 102: 465-81; Deshwal et al., Semin Respir Crit Care Med 2021, 42: 145-59).

**30:**

Treatments administered for hemoptysis via bronchoscopy may include topical vasoconstrictors, cautery, laser treatments, and balloon tamponade (Prey et al., Surg Clin North Am 2022, 102: 465-81; Atchinson et al., Am J Emerg Med 2021, 50: 148-55; Deshwal et al., Semin Respir Crit Care Med 2021, 42: 145-59).

**31:**

Topical hemostatic tamponade combines the application of a mechanical hemostatic agent and tamponade via bronchoscopy.

**32:**

The mainstay of treatment of patients with hemorrhagic shock is blood product transfusion (e.g., packed red blood cells). Despite the general principle of minimizing the use of intravenous fluids in patients with known hemorrhage, crystalloids are often administered until blood products are available, especially in patients who are hypotensive (e.g., mean arterial pressure less than 65). Vital sign changes may underestimate blood loss in patients. Patients with a 30%(0.30) loss of total blood volume may have little to no change in systolic blood pressure (SBP) and only demonstrate modest tachycardia (i.e., heart rate greater than 100). By the time an SBP has dropped to less than 100 mmHg, there may be as much as a 40%(0.40) loss of total blood volume.

**33:**

Fluid resuscitation involves the administration of repeated intravenous fluid boluses to acutely restore depleted intravascular volume. There is no evidence to support the use of colloids over crystalloids; studies have demonstrated that the primary choice for a resuscitation fluid is a balanced crystalloid solution, such as Lactated Ringer's (Myburgh, N Engl J Med 2018, 378: 862-3).
Bolus volumes for administration may be generally defined by the following:

- Fluid boluses of 250 to 1,000 milliliters in adults (Scales, Br J Nurs 2014, 23; Myburgh and Mythen, N Engl J Med 2013, 369: 1243-51; Vincent and De Backer, N Engl J Med 2013, 369: 1726-34)
- Fluid boluses of 10 to 20 milliliters per kilogram in pediatric patients. In some cases, boluses of 60 milliliters per hour or more may have to be administered in the first hour (Davis et al., Crit Care Med 2017, 45: 1061-93)

For patients with decreased cardiac reserve, significant renal dysfunction, or third-spacing, smaller volume boluses may be used (National Clinical Guideline Centre for Acute and Chronic Conditions, IV Fluids in Children: Intravenous Fluid Therapy in Children and Young People in Hospital. 2015. Reaffirmed 2020). The total number of fluid boluses required to restore intravascular volume and avoid fluid overload varies by patient (Joseph et al., J Trauma Acute Care Surg 2014, 76: 457-61; Holodinsky et al., Crit Care 2013, 17: R249; Kirkpatrick et al., Intensive Care Med 2013, 39: 1190-206). Dynamic measures such as bedside vena cava ultrasound and monitoring of urine output may assist in determining when volume resuscitation has been achieved (Kent et al., J Surg Res 2013, 184: 561-6; Zengin et al., Am J Emerg Med 2013, 31: 763-7; Weekes et al., Acad Emerg Med 2011, 18: 912-21).

**34:**
Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

**35:**
Depending on the location, bleeding in patients with hemophilia may be life-threatening and must be treated emergently. Intracranial hemorrhage is the leading cause of death in individuals with hemophilia and can occur spontaneously or after trauma. Bleeding into the calf or forearm may cause compartment syndrome and requires emergency treatment (Tebo et al., J Emerg Med 2020, 58: 756-66).

**36:**
One of the most serious complications in patients with hemophilia or von Willebrand disease is intracerebral hemorrhage. A computed tomography (CT) of the head should be obtained promptly in patients with a history of trauma, headache, altered mental status, or abnormal neurologic examination (Tebo et al., J Emerg Med 2020, 58: 756-66).

**37:**
Acute mental status change may include coma, stupor, obtundation, confusion, disorientation, delirium, or increasing lethargy (Shenvi et al., Ann Emerg Med 2020, 75: 136-45).

**38:**
Alteration in speech may include dysarthria (slurred or garbled speech), dysphasia (difficulty in comprehension or expression), or aphasia (full loss of language).

**39:**
Paresis refers to a condition of muscle weakness with a reduction in voluntary movement while paralysis refers to a complete loss of movement. These criteria include, but are not limited to, paresis or paralysis of the face as well as difficulty swallowing (i.e., dysphagia).
**Instruction**: Patients experiencing general weakness or inability to move limbs from a non-neurologic cause are excluded from these criteria.

**40:**
Sensory deficits seen in stroke or transient ischemia tend to be focal in nature and can include hemianesthesia, single limb anesthesia, facial hemianesthesia, or hypoesthesia (i.e., numbness).

**41:**
Visual field loss includes hemianopsia, quadrantanopia, or total blindness.

**42:**
Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage ($FiO_2$) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired $FiO_2$ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for $CO_2$ retention, where a precise inspired $FiO_2$ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired $FiO_2$. The following are estimates of $O_2$ delivered at the associated flow rates:
**LOW-FLOW SYSTEMS**
**Nasal cannula**

**Room air** 21%(*0.21*)
**1L** 24%(*0.24*) **4L** 36%(*0.36*)
**2L** 28%(*0.28*) **5L** 40%(*0.40*)
**3L** 32%(*0.32*) **6L** 44%(*0.44*)

**Simple oxygen masks**

- 35-50%(*0.35-0.50*) $FiO_2$ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of $CO_2$ in the mask

**Venturi masks**

- 24-50%(*0.24-0.50*) $FiO_2$ (4-12 L/min)

**Partial rebreathing masks**

- 40-70%(*0.40-0.70*) $FiO_2$ (6-10 L/min)

**Non-rebreathing masks**

- 60-80%(*0.60-0.80*) $FiO_2$ (minimum flow of 10 L/min)

**HIGH-FLOW SYSTEMS**
**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) $FiO_2$

**Heated, humidified high-flow concentrating nasal cannula (HFNC)**

- 24-100%(0.24-1.0) $FiO_2$

**43:**

High-flow nasal cannula (HFNC) delivers a fraction of inspired oxygen ($FiO_2$) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the $FiO_2$. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high-flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (i.e., face mask or nasal cannula with a flow rate of 15 L/min); however, it did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has been shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

**44:**

In addition to the potential need for supplementary oxygenation appropriate for management by mechanical ventilation, massive hemoptysis carries the risk of airway compromise. Mechanical ventilation may be utilized in hemoptysis for airway protection (Radchenko et al., J Thorac Dis 2017, 9: S1069-S86).

**45:**

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).
The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations.

InterQual® criteria do not differentiate between the different NIPPV modalities.
Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res 2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

**46:**

Patients with brisk or excessive uterine bleeding, symptoms of volume depletion, or symptoms of anemia should be considered for transfusion (Dyne and Miller, Emerg Med Clin North Am 2019, 37: 153-64).

**47:**

Contraindicated refers to **documented** clinical circumstances when a medication or treatment is not indicated based on the nature of the disease, comorbid condition, or a drug allergy or intolerance.

**48:**

Hormone therapy includes the administration of conjugated equine estrogen, combination estrogen and progestin, or progestin-only contraceptive medications (Borhart, In: Rosen's Emergency Medicine: Concepts and Clinical Practice, 10 edn. 2023:263-7; Dyne and Miller, Emerg Med Clin North Am 2019, 37: 153-64).

**49:**

Desmopressin is a synthetic analog of vasopressin which increases serum levels of factor VIII and von Willebrand factor in most patients with hemophilia A and von Willebrand's disease. Desmopressin should only be used for mild to moderate cases and is not recommended for severe disease. Tolerance may develop after 3 to 4 infusions of desmopressin. Factor levels should be monitored to confirm continuing response. All patients should undergo an initial trial prior to its use (Connell et al., Blood Adv 2021, 5: 301-25; Tebo et al., J Emerg Med 2020, 58: 756-66; Guidelines for emergency department management of individuals with hemophilia and other bleeding disorders. 2019).

**50:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.
**NOTE:** For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**51:**

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

**52:**

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

**53:**

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

**54:**

Repeat imaging includes:

- Computerized tomography (CT)
- CT angiography (CTA)

**55:**

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.
**Note**: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

**56:**

High flow nasal cannula (HFNC) delivers fraction of inspired oxygen ($FiO_2$) that is heated (up to 37 degrees Celsius) with 100%(1.0) relative humidity at flow rates up to 60 liters per minute, set independent from the $FiO_2$. The heated and humidified gas has been shown to improve comfort and aid in clearance of secretions. The high flow rates reduce anatomical dead space by flushing out carbon dioxide from the upper airways, leading to improved work of breathing. In addition, HFNC reduces the amount of ambient air a patient can breathe in, resulting in less oxygen dilution. A Cochrane review was conducted in 2021 to determine the effectiveness of HFNC compared to standard oxygen therapy and noninvasive positive pressure ventilation (NIPPV) in adult patients requiring respiratory support in the intensive care unit (ICU). The systematic review found that HFNC had less treatment failure compared to standard oxygen therapy (face mask or nasal cannula with a flow rate of 15 L/min); however, did not influence mortality rate or length of stay. When HFNC was compared to NIPPV, which included bilevel positive airway pressure (BiPAP) or continuous positive airway pressure (CPAP), there was no difference in treatment failure defined as intubation or re-intubation, mortality, or length of stay. In conclusion, HFNC has shown to have less treatment failure compared to standard oxygen therapy and may be an alternative to NIPPV in patients treated for acute hypoxic respiratory failure (Lewis et al., Cochrane Database Syst Rev 2021, 3: Cd010172).

**57:**

Respiratory monitoring includes an evaluation of the patient's respiratory rate, heart rate, oxygen saturation, or work of breathing. Within 90 minutes of high-flow nasal cannula (HFNC) initiation, the patient's respiratory status should improve, indicating adequate respiratory support is being provided. If clinical improvement is not achieved within this time frame, additional respiratory support may be indicated (Lee et al., Intensive Care Med 2013, 39: 247-57).

**58:**

**Instruction:** Application of this criteria point is appropriate for patients post embolization due to spontaneous hematoma (e.g., retroperitoneal hematoma, intraperitoneal hematoma, rectus sheath hematoma).

**59:**

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

**60:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient’s need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.
For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.
**Note:** Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status;** those who do not meet criteria would be appropriate for secondary review.
(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

**61:**

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

**62:**

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

**63:**

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

**64:**

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

**65:**

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

**66:**

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

**67:**

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

**68:**

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs
- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

**69:**

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

**70:**

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

**Evaluation and Treatment**

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

**71:**

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

**72:**

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term complications
- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

**73:**

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

**74:**

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

**75:**

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

**76:**

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.
The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

**77:**

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

**78:**

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

**79:**

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

**80:**

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)

- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

**81:**

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.